WEEK 2 · SUNDAY

Loss, probability and matrices

How a model learns from its mistakes.

AI for Beginners · Lesson 2, Part B



TODAY'S PLAN

Yesterday's tools, put to work

YESTERDAY IN ONE LINE

The **gradient** points uphill. To cut the error, walk the **opposite** way.

Today we find out what "the error" really is, and then take the walk.

1. **Loss and gradient descent**
   Measure the mistake, nudge, repeat.
2. **Probability**
   How AI deals with being unsure.
3. **Matrices**
   The tables of numbers inside every model.

LOSS

Loss: a score for how wrong we are

A model makes a guess. We compare it with the real answer and squash the gap into **one number**: the **loss**.

How to read it

Small loss = good guess.
Zero loss = perfect.

Training means: make the loss as small as possible.



LOSS

A popular loss: mean squared error

1. **Subtract.** Error = guess − truth
2. **Square.** No negatives, big misses hurt more
3. **Average** over all examples

House	True price ($k)	Model's guess	Error	Error squared
A	100	110	+10	**100**
B	150	140	−10	**100**
C	200	220	+20	**400**

(100 + 100 + 400) ÷ 3 = 200 That average is the loss. Lower is better.

OUR TINY MODEL

One input, one knob, one guess



Start with w = 1

Guess: $\hat{y} = 1 \times 2 = \mathbf{2}$

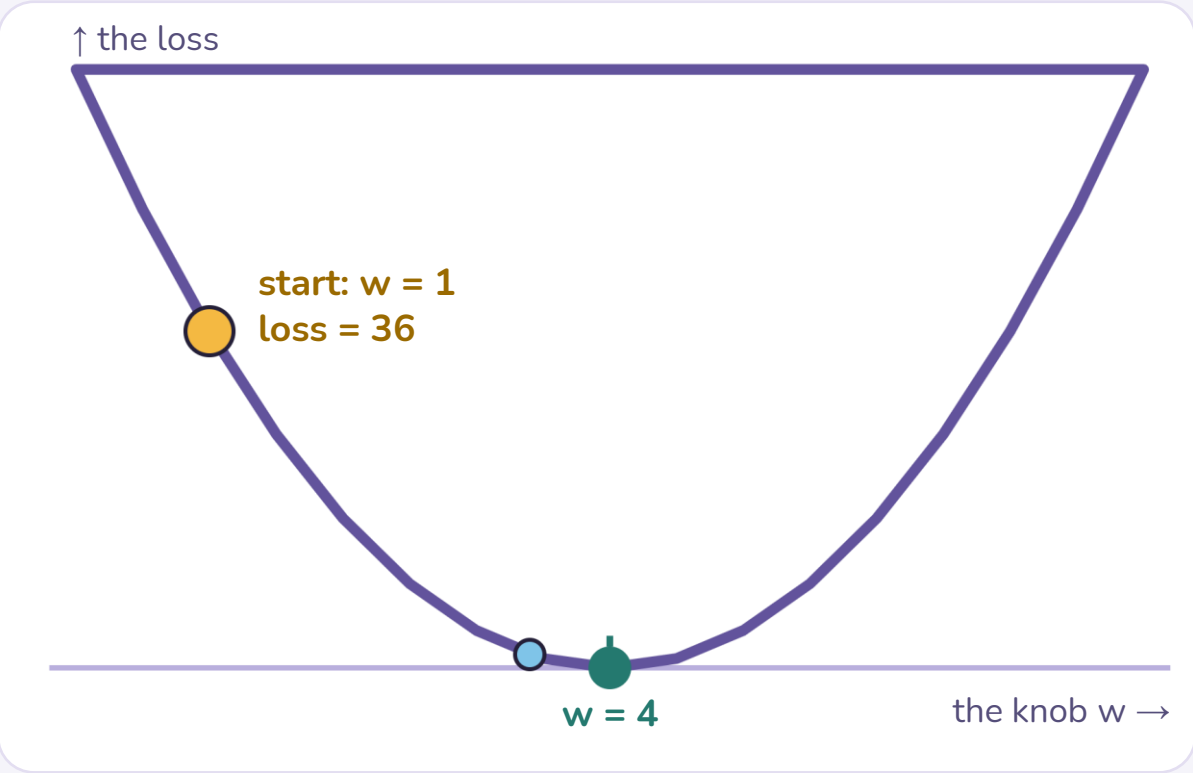
Loss: $(2 - 8)^2 = \mathbf{36}$

The job

Find the w that makes the guess land on 8. Do it by letting the **loss** and the **slope** guide us.

THE LOSS LANDSCAPE

Try every knob setting: you get a bowl



Plot the loss for **every** possible w. Far from 4, the loss is high. At 4, it hits zero.

The picture to keep

Training is a ball rolling down this bowl. The **slope** tells it which way is down.

For this tiny model the bowl is a simple curve. For ChatGPT-sized models it lives in billions of dimensions. The idea is the same.

THE SLOPE OF THE LOSS

Chain rule: how does the loss react to w?



Loss reacts to the guess

$dL/d\hat{y} = 2(\hat{y} - y)$

Guess reacts to the knob

$d\hat{y}/dw = x$

Multiply: $dL/dw = 2(\hat{y} - y) \cdot x$

At w = 1: 2 × (2 − 8) × 2 = **−24**. Negative slope: raising w lowers the loss.

GRADIENT DESCENT

The update rule: one step downhill

wnew = w – lr × slope

Move against the slope. Steeper slope means a bigger step.

lr = learning rate: the size of our stride. We pick it ourselves.

Here: lr = 0.1

Step 1, by hand

w = 1 – 0.1 × (–24)

w = 1 + 2.4 = **3.4**

New guess: 3.4 × 2 = **6.8**

New loss: (6.8 – 8)² = **1.44**

Loss: 36 → 1.44 in one step!

GRADIENT DESCENT

Repeat: watch the loss melt away

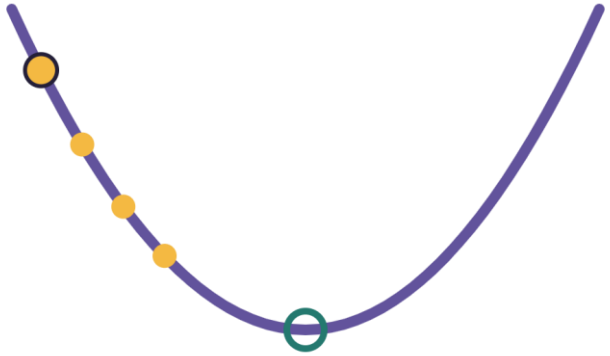
Step	Knob w	Guess	Loss	Slope
0	1	2	**36**	−24
1	3.4	6.8	**1.44**	−4.8
2	3.88	7.76	**0.0576**	−0.96
3	3.976	7.952	**0.0023**	−0.19
4	3.995	7.990	**0.0001**	−0.04

Gentle landing: as the bowl flattens, the slope shrinks, so the steps shrink too.

The model **discovered w ≈ 4** without anyone telling it.

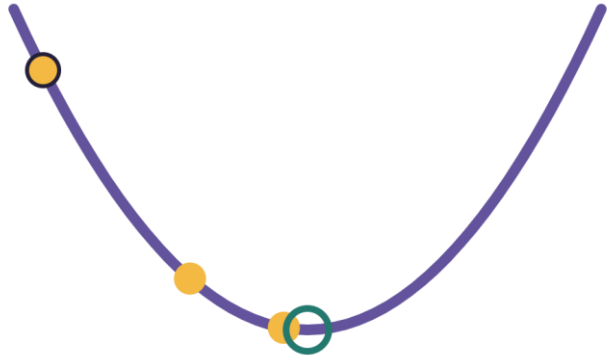
GRADIENT DESCENT

The learning rate: how big is each stride?



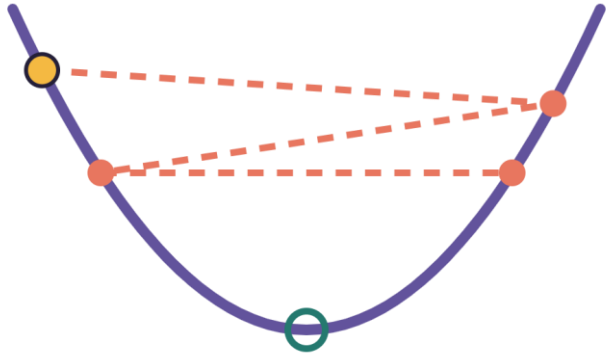
Too small

Safe, but painfully slow. It may take forever to reach the bottom.



Just right

Big steps at first, smaller as it nears the bottom. Lands smoothly.



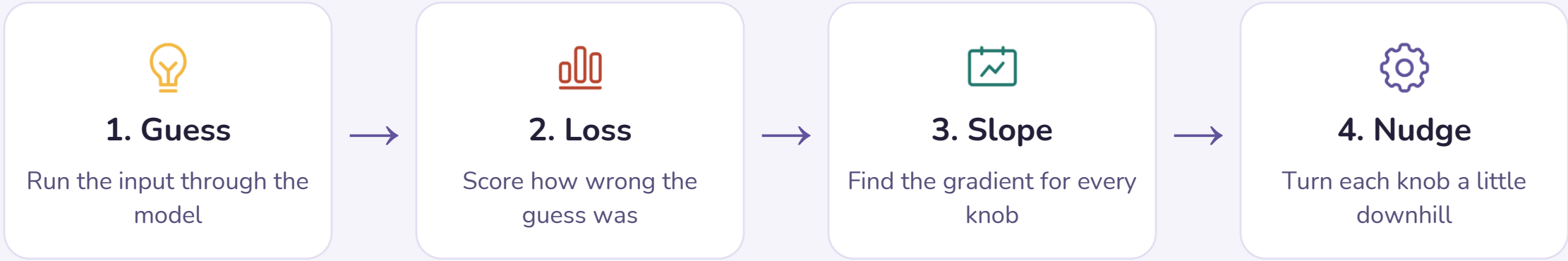
Too big

Overshoots and bounces from side to side. The loss may even go up.

Choosing lr is part of the craft of AI. Engineers try a few values and watch the loss.

THE BIG PICTURE

This loop is how models learn



↺ **Then go back to step 1.** Repeat until the loss stops shrinking.

A chatbot is trained with this same loop. More knobs, more data, many more rounds, but **the same four steps**.

QUICK CHECK

Which way should the knob move?

A model's knob is at **w = 5**. The slope of the loss there is **+3** (positive).

To lower the loss, what should we do to w?

A Increase it

B Decrease it

C Leave it alone

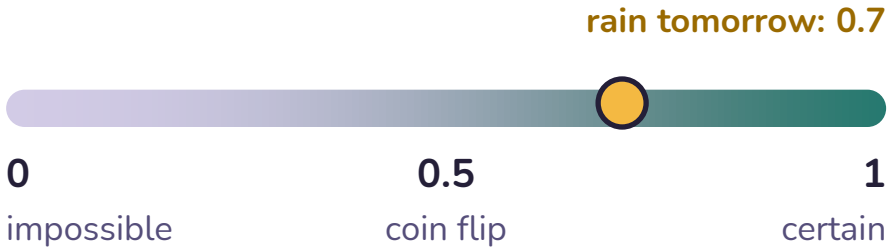
Hint: the update rule is w − lr × slope.

PROBABILITY

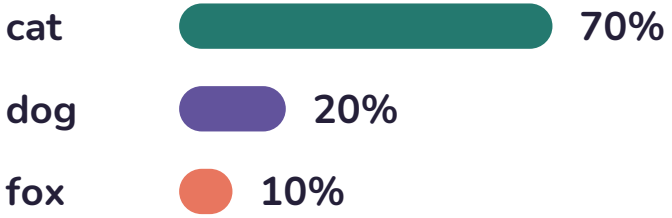
AI is never 100% sure, and that is fine

A scale from 0 to 1

Probability measures how likely something is. 0.5 means 50%.



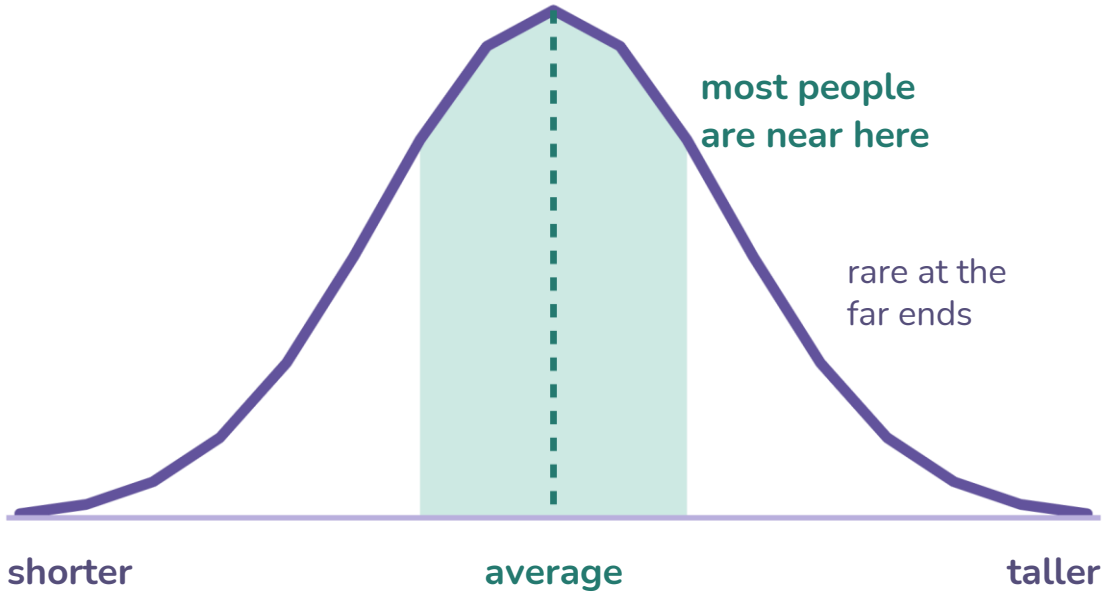
What the model says about a photo



Rule: all the options always add up to 100%.

PROBABILITY

Data has a shape: the bell curve



Measure the heights of 1,000 students. Most land **near the average**; very few are extremely short or tall. Plot it and you get a **bell**.

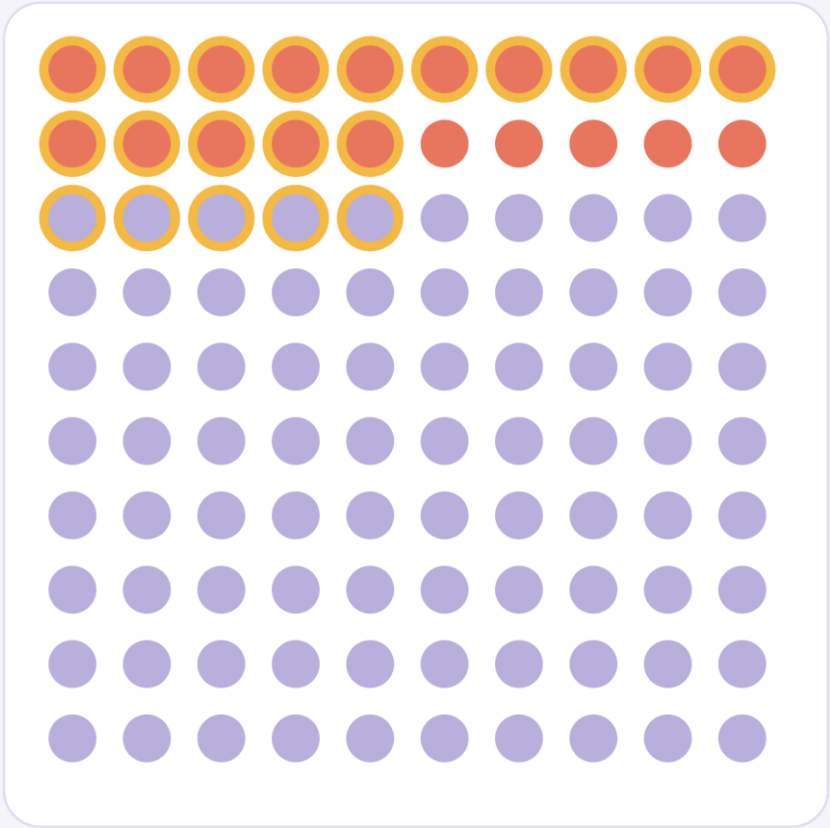
More bell-shaped things

exam scores, delivery times, shoe sizes, small measurement errors

Why AI cares: a value far from the middle is **surprising**. Models use that to spot fraud, faults and oddities.

PROBABILITY

Evidence changes the odds: a spam filter



spam normal contains "free"

Before reading anything

20 of 100 emails are spam: **20%**

Now see the word "free"

20 emails have it. 15 of them are spam: 15 ÷ 20 = **75%**

The evidence moved the belief from **20%** to **75%**. This is **Bayes' idea**: update what you believe when you see new clues.

PROBABILITY IN MODELS

Softmax: turning scores into probabilities

A model first produces raw **scores**. Softmax converts them into **percentages that add to 100%**.

Option	Score	escore	Probability
cat	2.0	7.39	**66%**
dog	1.0	2.72	**24%**
fox	0.1	1.11	**10%**

Two steps: make every score positive (escore), then divide each by the total, 11.22.

The biggest score always wins the biggest share. **ChatGPT uses this to pick words.**

LOSS FOR PROBABILITIES

Cross-entropy: punish confident mistakes

The photo really is a **cat**. How much loss does the model get, depending on the probability it gave to "cat"?

The model says…	P(cat)	Loss
Confident and right	0.9	0.11
Unsure	0.5	0.69
Mostly wrong	0.1	2.30
Confident and wrong	0.01	4.61

loss = −ln(P of the right answer)

Being **sure and wrong** is punished far more than being unsure.

MATRICES

A matrix is just a table of numbers

Data is a table

Each row is one house. Each column is one fact about it.

size m²	rooms	age
80	3	10
120	4	5
45	1	30

3 rows × 3 columns = a 3×3 matrix

A picture is a table too

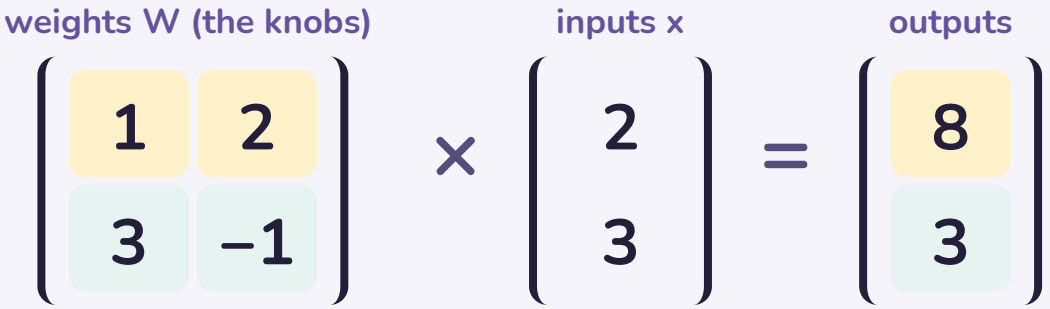
Every pixel is a number: 0 is white, 9 is dark.

0	0	9	0	0
0	0	9	0	0
9	9	9	9	9
0	0	9	0	0
0	0	9	0	0

Models cannot see houses, photos or words. They only see **tables of numbers**.

MATRICES

One layer of a network is a matrix times a list



Top row × inputs:
1×2 + 2×3 = 8

Bottom row × inputs:
3×2 + (−1)×3 = 3

Each row multiplies and adds. Real models do this with **millions of numbers** at once, which is why AI loves fast chips called **GPUs**.

RECAP

Four things to remember

Loss

One number for how wrong the model is. Lower is better.

Gradient descent

Guess, score, find the slope, nudge downhill. Repeat.

Probability

Models give odds, not certainties. Evidence updates the odds.

Matrices

Tables of numbers. A layer multiplies them by the inputs.

Whole story in one line: a model is a pile of numbers, tuned by walking downhill on its mistakes.

Next week: how to feed real data to a model, and the classic ML algorithms.

BEFORE NEXT WEEK

Try it yourself

1. Our neuron guesses $\hat{y} = 5$ when the truth is $y = 8$. What is the squared-error loss?

2. Compute **[[2, 0], [1, 1]] × [3, 4]** using the row rule.

3. In your own words: why is a model that is **confident and wrong** worse than one that is unsure?

Bring your answers next Saturday. We start with them.